

Nonparametric estimation of causal parameters with binary outcomes

Sangmyung Ha

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1 Unconfoundedness

Suppose we have two binary potential outcomes $Y(1), Y(0)$ given by

$$\begin{aligned} Y(1) &= 1\{Y(1)^* \geq 0\} \\ Y(0) &= 1\{Y(0)^* \geq 0\} \end{aligned}$$

where $Y(1)^*, Y(0)^*$ are two potential latent utilities associated with the choice $Y(1) = 1$ if treated and the choice $Y(0) = 1$ if untreated. Without loss of generality, suppose the potential latent utilities are generated as

$$\begin{aligned} Y(1)^* &= m(1, V) - \varepsilon(1) \\ Y(0)^* &= m(0, V) - \varepsilon(0) \end{aligned}$$

where $m(1, V)$ and $m(0, V)$ are nonparametric systematic components of $Y(1)^*$ and $Y(0)^*$ respectively, and the cumulative distribution functions of the error terms $\varepsilon(1), \varepsilon(0)$ are given by

$$\begin{aligned} \mathbb{P}\{\varepsilon(1) \leq z\} &= F_1(z) \\ \mathbb{P}\{\varepsilon(0) \leq z\} &= F_0(z) \end{aligned}$$

We see that the conditional choice probabilities for each potential outcome are given by

$$\begin{aligned} \mathbb{P}\{Y(w) = 1 | V = v\} &= \mathbb{P}\{Y(w)^* \geq 0 | V = v\} \\ &= F_w(m(w, v)) \end{aligned}$$

for $w = 1, 0$.

Suppose that the unconfoundedness assumption holds so that the assignment to treatment $W \in \{1, 0\}$ can be viewed as random, conditional on the observed V .

$$\{Y(1), Y(0)\} \perp W | V \tag{1}$$

Given V , the randomness in $Y(1)$ and $Y(0)$ comes only from $\varepsilon(1)$ and $\varepsilon(0)$ so that the unconfoundedness assumption implies

$$\{\varepsilon(1), \varepsilon(0)\} \perp W | V$$

Under the unconfoundedness assumption, we have

$$\begin{aligned}
P\{Y = 1|V = v, W = w\} &= P\{Y(w) = 1|V = v, W = w\} \\
&= P\{Y(w)^* \geq 0|V = v, W = w\} \\
&= P\{\varepsilon(w) \leq m(w, v)|V = v, W = w\} \\
&= P\{\varepsilon(w) \leq m(w, v)|V = v\} \\
&= F_w(m(w, v))
\end{aligned}$$

for $w = 1, 0$, where we've used the unconfounded assumption on the fourth equality.

Here, the nonparametric systematic component $m(w, v)$ and the cumulative distribution of the error terms F_w are not separately identified. This implies that as long as we have a fully nonparametric specification of $m(w, v)$, we may assume F_1 and F_0 are logistic distributions without loss of generality. To see this, let $Y(1)^*, Y(0)^*$ be generated as

$$\begin{aligned}
Y(1)^* &= \tilde{m}(1, V) - \tilde{\varepsilon}(1) \\
Y(0)^* &= \tilde{m}(0, V) - \tilde{\varepsilon}(0)
\end{aligned}$$

where $\tilde{m}(1, V)$ and $\tilde{m}(0, V)$ are another nonparametric systematic components of $Y(1)^*$ and $Y(0)^*$ respectively, and the error terms both follow logistic distribution

$$\mathbb{P}\{\varepsilon(1) \leq z\} = \mathbb{P}\{\varepsilon(0) \leq z\} = \frac{1}{1 + e^{-z}}$$

Here, if we let $\tilde{m}(w, v) = -\log(F_w(m(w, v))^{-1} - 1)$, the conditional choice probability is given by

$$\begin{aligned}
P\{Y = 1|V = v, W = w\} &= P\{Y(v) = 1|V = v, W = w\} \\
&= P\{Y(v)^* \geq 0|V = v, W = w\} \\
&= P\{\tilde{\varepsilon}(v) \leq \tilde{m}(w, v)|V = v, W = w\} \\
&= P\{\tilde{\varepsilon}(v) \leq \tilde{m}(w, v)|V = v\} \\
&= \frac{1}{1 + e^{-\tilde{m}(w, v)}} \\
&= \frac{1}{1 + e^{\log(F_w(m(w, v))^{-1} - 1)}} \\
&= F_w(m(w, v))
\end{aligned}$$

Thus, the conditional choice probability of the observables can always be written as

$$P\{Y = 1|V = v, W = w\} = \frac{1}{1 + e^{-m(w, v)}}$$

and the conditional choice probability of the potential outcomes can also be written as

$$P\{Y(w) = 1|V = v\} = \frac{1}{1 + e^{-m(w, v)}}$$

for $w = 1, 0$ for some nonparametric function $m(w, v)$. This implies that given the data $\{Y_i, V_i, W_i\}_{i=1}^n$, we may estimate the nonparametric function $m(w, v)$ using the likelihood function

$$\ell(y, v, w) = -y \log(1 + e^{-m(w, v)}) - (1 - y) \log(1 + e^{m(w, v)})$$

and use the estimated $\hat{m}(w, v)$ to obtain causal parameter of interest.

2 Models for the nonparametric systematic part $m(w, v)$

1. We may use a class of simple functions to model $m(w, v)$ with the binary splitting algorithm for estimation. This corresponds to fitting a decision tree classifier with the cross entropy loss function.
2. We may specify $m(w, v)$ as an element of RKHS associated with a given positive definite kernel.
3. We may specify $m(w, v)$ using a neural network. This corresponds to estimating a neural network with a sigmoid output layer and cross-entropy loss function.

3 Estimation of Causal Parameters

Recall that the conditional choice probability for $Y(1)$ and $Y(0)$ are given as

$$P\{Y(w) = 1|V = v\} = \frac{1}{1 + e^{-m(w, v)}}$$

for $w = 0, 1$. The causal parameter of interest is the conditional partial effect defined as

$$\begin{aligned}\tau(v) &= P\{Y(1) = 1|V = v\} - P\{Y(0) = 1|V = v\} \\ &= \frac{1}{1 + e^{-m(1, v)}} - \frac{1}{1 + e^{-m(0, v)}}\end{aligned}$$

if W is binary and

$$\begin{aligned}\tau(w, v) &= \frac{\partial P\{Y(w) = 1|V = v\}}{\partial w} \\ &= \frac{e^{-m(w, v)}}{(1 + e^{-m(w, v)})^2} \frac{\partial m(w, v)}{\partial w}\end{aligned}$$

if W is continuous.¹ For binary W , the estimation of conditional partial effect is straightforward

$$\hat{\tau}(v) = \frac{1}{1 + e^{-\hat{m}(1, v)}} - \frac{1}{1 + e^{-\hat{m}(0, v)}}$$

For continuous W , the conditional partial effect may be estimated by

$$\hat{\tau}(w, v) = \frac{e^{-\hat{m}(w, v)}}{(1 + e^{-\hat{m}(w, v)})^2} \frac{\partial \hat{m}(w, v)}{\partial w}$$

as long as $\frac{\partial \hat{m}(w, v)}{\partial w}$ exists. If $\hat{m}(w, v)$ is estimated using either RKHS or neural network, $\frac{\partial \hat{m}(w, v)}{\partial w}$ exists. If $\hat{m}(w, v)$ is estimated using random forest, $\frac{\partial \hat{m}(w, v)}{\partial w}$ does not exist. One potential remedy for this is to use a structured random forest. In other words, if we model $m(w, v)$ as having either a partially linear or functional-coefficient form, we have

$$m(w, v) = \beta'w + f(v)$$

¹Assuming that $Y(W)$ exists.

or

$$m(w, v) = \beta(v)'w$$

so that the partial derivative $\frac{\partial \hat{m}(w, v)}{\partial w}$ always exists as either $\hat{\beta}$ or $\hat{\beta}(v)$. After estimating the conditional partial effect, we may estimate the average partial effect as

$$\hat{\mathbb{E}}[\hat{\tau}(V)] = \frac{1}{n} \sum_{i=1}^n \hat{\tau}(V_i), \quad \text{or} \quad \hat{\mathbb{E}}[\hat{\tau}(W, V)] = \frac{1}{n} \sum_{i=1}^n \hat{\tau}(W_i, V_i)$$

and the average partial effect on the treated as

$$\hat{\mathbb{E}}[\hat{\tau}(V)|W = 1] = \frac{1}{|I_1|} \sum_{i \in I_1} \hat{\tau}(V_i), \quad \text{or} \quad \hat{\mathbb{E}}[\hat{\tau}(W, V)|W = 1] = \frac{1}{|I_1|} \sum_{i \in I_1} \hat{\tau}(1, V_i)$$

where I_1 is the index set for the treated observations.

4 Considerations

1. Here, we use logistic distribution for the error terms without loss of generality because given the conditional choice probability

$$P\{Y = 1|V = v, W = w\} = F_w(m(w, v))$$

$m(w, v)$ and F_w are not separately identified. However, under the assumptions given in Yan (2024), it is possible to separately identify $m(w, v)$ and F_w where both are estimated nonparametrically. However, doing this would introduce additional computational burden while also unnecessarily complicating the model. If the causal parameter of interest is the conditional partial effect or average partial effect, there is no need to separately identify $m(w, v)$ and F_w .

2. If we add structure to the nonparametric systematic part $m(w, v)$, then assuming logistic distribution for the error term comes at a loss of generality. However, this might be beneficial for interpretation as well as for computation of partial effects for continuous treatments.